

■ Scientific Justification (required for all)

Does circumnuclear dust remember an AGN’s accretion history? Large luminosity changes in quasars and other active galactic nuclei provide timed experiments on circumnuclear dust. The relevant population includes strongly variable quasars [24] and changing-look AGN [1]. Existing grains heat or cool as the changed radiation reaches them; grains may also be destroyed or form at new locations. **Can delayed illumination of a fixed dust distribution explain the infrared spectra through brightening, fading and recovery, or must the emitting distribution evolve?** We use measured luminosity histories to test this question. The answer determines how infrared spectra trace an AGN’s power and dust covering factor during variability.

The equilibrium sublimation radius scales approximately as $L^{1/2}$ [2,3], but the emitting distribution need not adjust immediately. Brightening tests whether increased heating suffices or the inner dust changes; after fading, surviving dust cools and newly formed inner dust may restore a hotter component. More distant dust echoes earlier illumination. We propose joint MIRI/MRS and NIRSpec IFU spectroscopy of an 18-object working sample to compare the hot and warm components against each source’s preceding history, with repeated tests across different nuclei. Figure 1 shows P1823 in a high optical state, with a long spectral baseline and a WISE decline followed by recovery.

From infrared variability to dust physics. WISE observations establish infrared responses to changing-look events [4–6], but a change in infrared colour can reflect temperature, emitting area, extinction or host dilution. Individual reverberation studies show why a broader test is needed. In Mrk 590, the inner dust radius became smaller following a luminosity decline, implying replenishment within a few years [7]. In NGC 4151, the hottest dust remained at roughly the same radius over decades, while the warm emission had a much longer response [8]. These results motivate testing contrasting histories.

JWST supplies the missing spectral information. MRS spectra reveal substantial diversity beyond current dust models [9], including a distinctive silicate spectrum in the recently activated nucleus Ansky [10]. We test the same physical model against spectra and preceding histories. **The gain is a constrained comparison of delayed heating and dust evolution across known histories.**

What JWST adds to SPHEREx. SPHEREx already provides the near-infrared baseline: **0.75–5 μm spectroscopy [25].** P1823 has three dated epochs (Figure 1). The principal wavelength extension is MIRI/MRS, which measures the warm continuum, accessible silicate features, aromatic bands and ionic lines beyond 5 μm . These constrain the warm dust and host contribution against which we interpret the hot component. NIRSpec largely overlaps SPHEREx in wavelength; its added value is $R \simeq 1000$ line and continuum detail, spatial information to separate measurable host emission, and a nuclear spectrum near the MIRI epoch. Broad Pa α constrains the nuclear state and reddening. The existing SPHEREx spectra remain independent constraints at their actual dates.

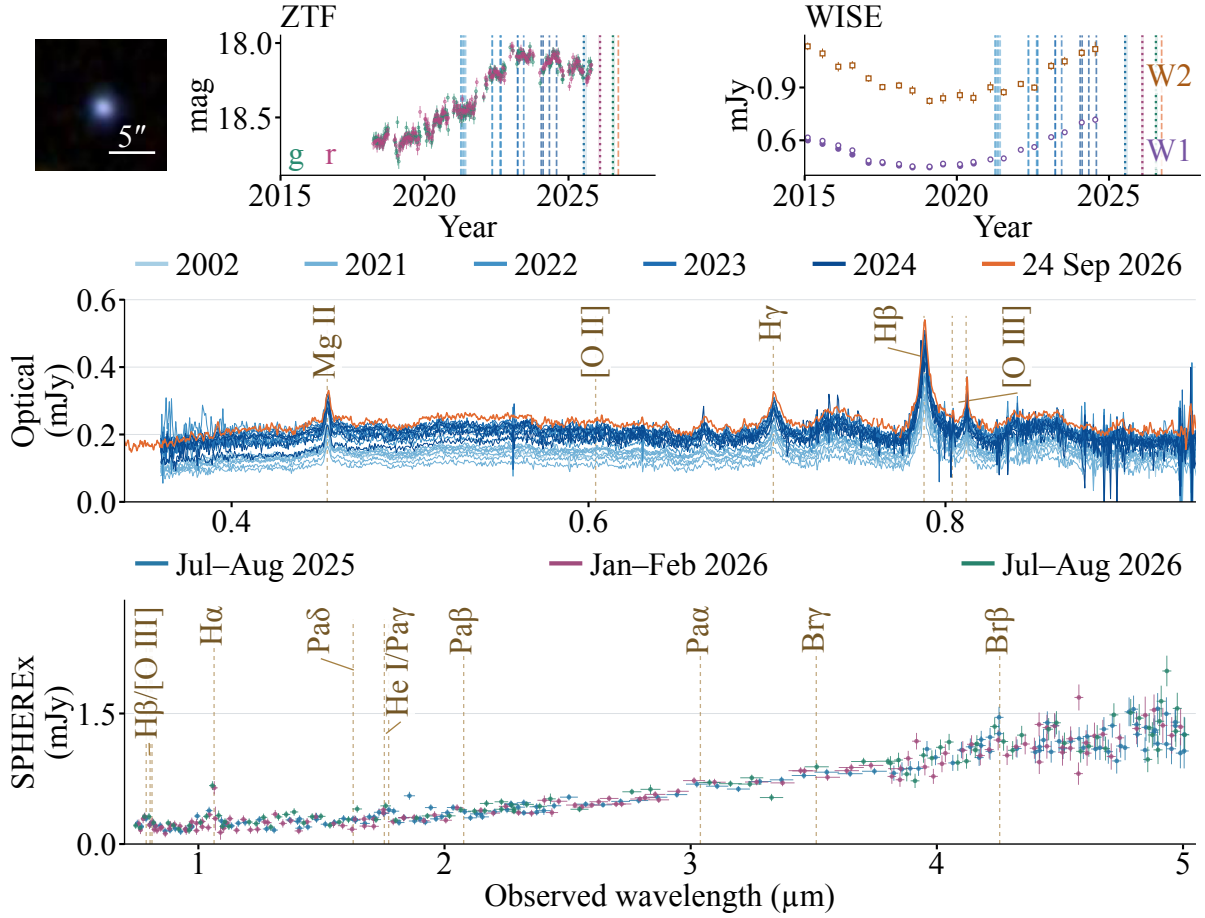


Figure 1: **P1823 connects a measured luminosity history to the infrared spectrum.** The quasar ($z = 0.620$) is in a high optical state; WISE records an earlier decline and recovery. Top: SDSS cutout (5'' bar), ZTF g, r and WISE W1 (purple circles), W2 (brown squares) histories from 2015. Middle: 21 optical epochs. Bottom: three SPHEREx visits. Date keys above both spectra match the light-curve markers: dashed lines mark optical epochs; dotted lines mark SPHEREx median dates, with shading spanning each visit. The 2002 spectrum predates the displayed time window. Spectral axes use observed μm and F_{ν} in mJy; WISE fluxes are in mJy. Gold markers show expected emission-line positions, not fitted detections.

Can changing illumination explain the dust spectrum? Figure 1 supplies the optical history and dated near-infrared spectra. MIRI adds the longer-wavelength dust spectrum. We first ask whether a fixed dust distribution, heated by the measured luminosity history, can reproduce these observations at their actual dates, including light-travel delays, host emission and reddening. We then allow the amount or location of inner dust to change and test whether the data require that extra freedom. Agreement with fixed dust places limits on structural changes; a persistent mismatch motivates testing dust evolution.

■ Description of Observations (required for all GO and Survey)

Targets and wavelength coverage. The working sample contains ten infrared-selected variable candidates [16] and eight catalogue CLAGN [11,12,14,15]. Seventeen have $z < 0.3$, placing both silicate peaks within MRS; P1823 ($z = 0.620$) supplies a well-sampled optical/SPHEREx history and the $9.7\ \mu\text{m}$ feature at $15.7\ \mu\text{m}$. Its $18\ \mu\text{m}$ peak lies beyond MRS. The primary diagnostics do not depend on channel 4C [17]. Archival total-source W3 and W4 fluxes span approximately 1.4–24 and 3.3–75 mJy; nuclear fluxes must be established for exposure design. P8548’s W4 S/N is only 2.5, so its historical baseline is weak.

Selecting histories that test different response times. The relevant clock is the rest-frame time since a luminosity change, $\Delta t = (t_{\text{JWST}} - t_{\text{change}})/(1+z)$, compared with the expected rest-frame hot-dust delay, τ_{dust} . These delays depend on luminosity, approximately as $L^{1/2}$ [3,26]; equal calendar ages do not imply equal response phases. We bracket changes with optical photometry and spectra, and estimate lag ranges from measured luminosities or reverberation constraints where available. Uncertain event dates, host subtraction and lag scatter enter the comparison.

We prioritize complementary histories: rising sources test prompt heating against the slower warm response; sustained faint states test cooling and possible inner-dust replenishment; recovery tests whether the infrared spectrum depends on earlier illumination as well as current brightness. For a decline followed for several hot-dust delays, a remaining hot-component mismatch is harder to attribute to the initial echo alone. The warm component can still retain an older history [7,8]. Each source is tested against its own dated baseline, with luminosity and host differences included in comparisons between sources.

The saved light curves already provide contrasting examples: P1823 brightens over 2019–2023; P11530 fades over 2018–2021 and partly recovers in 2022–2023; P9694 brightens between 2021 and 2022 and remains brighter through 2025. Additional histories include optical decline through 2025, recovery after earlier fading, and a reversal in 2025. These trend intervals will probe multi-year aftermaths by Cycle 6; coverage within one hot-dust delay is not established. Final membership and history assignments require fitted event intervals and lag ranges. Broad-line appearance or disappearance is not required.

Observing at an informative epoch. Cycle 6 nominally runs from July 2027 to June 2028 [27]. Selection therefore evaluates each history at the allowed JWST dates. We favor sources whose model comparison remains informative across broad visibility windows, rather than depending on catching an unpredictable peak. Contemporary optical coverage is needed to connect the saved light curves, mostly ending in 2025, to JWST. Earlier WISE and available SPHEREx epochs constrain the preceding dust response; they are not assumed to represent the JWST state.

Why eighteen sources? The design targets roughly six sources in each broad history family—rising, fading and recovering—to distinguish recurring behaviour from individual peculiarities. This is an allocation goal, not yet a validated division of the working list. For independent measurements with common within-family scatter s , six objects give a mean uncertainty of $0.41s$ and a difference-of-means uncertainty of $0.58s$ between two families. These scalings motivate replication; they are not a target-specific power forecast. We will test sensitivity to individual objects and include unequal errors, intrinsic scatter and shared model uncertainties. Inference

concerns this deliberately selected sample; population-wide rates would require a selection function.

MIRI/MRS. One visit per target uses all three grating settings, target acquisition and a four-point dither, covering approximately $4.9\text{--}27.9\ \mu\text{m}$ [17,18]. FASTR1 is the starting readout configuration. The planning allocation includes dedicated skies; target-specific nuclear fluxes and backgrounds determine the integration lengths. Exposure design is driven by the nuclear continuum and the $18\ \mu\text{m}$ feature, with goals of $S/N \geq 50$ per $R = 100$ continuum bin near rest 5 and $12\ \mu\text{m}$ and ≥ 30 across the longer-wavelength feature. These are requirements to verify with target-specific ETC calculations.

NIRSpec and timing. G235M/F170LP and G395M/F290LP provide $1.66\text{--}5.27\ \mu\text{m}$ spectra at $R \simeq 1000$ [19], overlapping the SPHEREx baseline. Their purpose is to measure the nuclear continuum and broad $\text{Pa}\alpha$ near the MIRI epoch, with spectral and spatial detail for constraining reddening and measurable host emission. The working allocation is 300 s per grating in total across four dithers, with a continuum S/N goal of at least 50 per $R = 100$ bin. Readout, saturation and nuclear flux assumptions require ETC validation.

One epoch per instrument tests each source against its preceding history. The working maximum separation of MIRI and NIRSpec is 90 observer-frame days; each target’s predicted near-infrared evolution must be checked against the precision needed to distinguish the models, and the interval tightened before submission where necessary. Actual dates enter the fit. Supporting optical spectroscopy is planned near the JWST visits. Repeat JWST epochs would require a predicted detectable change over an accessible interval; none are included in the present allocation.

Resources and feasibility. The eighteen-target design has a **planning allocation of approximately 60–83 h** for eighteen MRS visits with dedicated-sky allowances and eighteen NIRSpec visits. This is a Medium program at the planning stage. Target-specific ETC/APT calculations must validate the total using conservative nuclear spectra, host backgrounds and current throughput. Wavelength-dependent calibration errors enter separately from statistical noise [20].

Using the archival baseline correctly. The 2010 W3/W4 fluxes constrain earlier emission in large apertures. We predict them by integrating redshifted nuclear-plus-host models through the WISE bandpasses, including colour corrections and uncertainty beyond MRS coverage. Archival imaging constrains host light outside the MRS field; an uncertain nuclear baseline becomes an interval or upper bound. Each target has its own host and aperture model, checked against imaging and spatially resolved host emission where measurable.

Joint spectral and temporal inference. We will decompose each cube into a point source and extended emission [9], then jointly fit continuum, silicates, aromatic bands, H_2 and ionic lines [21]. Aromatic bands and [Ne II] constrain host contamination; [Ne V] and [Ne VI] measurements or limits constrain the hard ionizing component. The torus remains spatially unresolved. Optical decomposition constrains the illumination history, marginalizing over unobserved intervals and the conversion to heating luminosity. WISE and available SPHEREx spectra are fitted as reprocessed emission at their actual dates.

The fixed-distribution model varies illumination with retarded time while keeping geometry and grain properties constant, spanning clumpy and disk-plus-wind geometries [22,23]. The evolving model allows the inner emitting distribution to change while treating illumination, extinction and host emission in the same way. Silicate absorption alone is not a unique test of obscuration. Model comparisons include wavelength-dependent calibration covariance and alternative host decompositions; simulated observations establish which changes are distinguishable at the achieved precision. Residuals shared across the sample indicate shortcomings in the model family. Residuals associated with luminosity history motivate additional dust evolution.

A common diagnostic across histories. The primary ensemble diagnostic is the logarithmic residual in the hot-to-warm dust-continuum luminosity ratio (rest 2–5 to 8–13 μm) relative to the fixed-distribution prediction at the observed dates. The decomposition separates dust from direct continuum, lines and host light. We compare rising and fading histories first, with recovery as a secondary comparison, and model the residual against continuous $\Delta t / \tau_{\text{dust}}$. Class labels do not replace the full illumination history. Luminosity and host contribution enter as covariates; source-by-source fits and removal of individual objects test whether an ensemble result is driven by one unusual nucleus. Simulated spectra and histories must establish the detectable effect size and final statistical power before submission.

Supplemental Information

Special requirements. The working design groups MIRI and NIRSpec for each target within 90 observer-frame days. Target-specific variability modelling will verify or tighten that limit before submission. Any narrower scientifically necessary date window must also be justified and entered in APT [27]. Visibility and schedulability require final APT checks. Exact dates and intervening variability enter the modelling. No coordinated parallels or joint-facility time are requested. Supporting optical observations are planned through an existing program.

Duplications. The working list requires a target-by-target archive and APT duplication check, including approved observations. Archival infrared spectra, where available, will be included at their original dates and apertures.

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Generative AI disclosure. Earlier drafts used OpenAI Codex (GPT-6; build identifier not exposed) and Claude Fable 5.1 (Anthropic), as recorded in the draft history, for literature searches, text, figures and preliminary observing estimates. OpenAI Codex assisted with the present revision on 2026 September 25–26 by comparing drafts, checking the model implementation and selected sources, revising text and figures, and checking the compiled document. Target histories, references and observing configurations require final verification by the proposers, who retain responsibility for the submitted content.

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